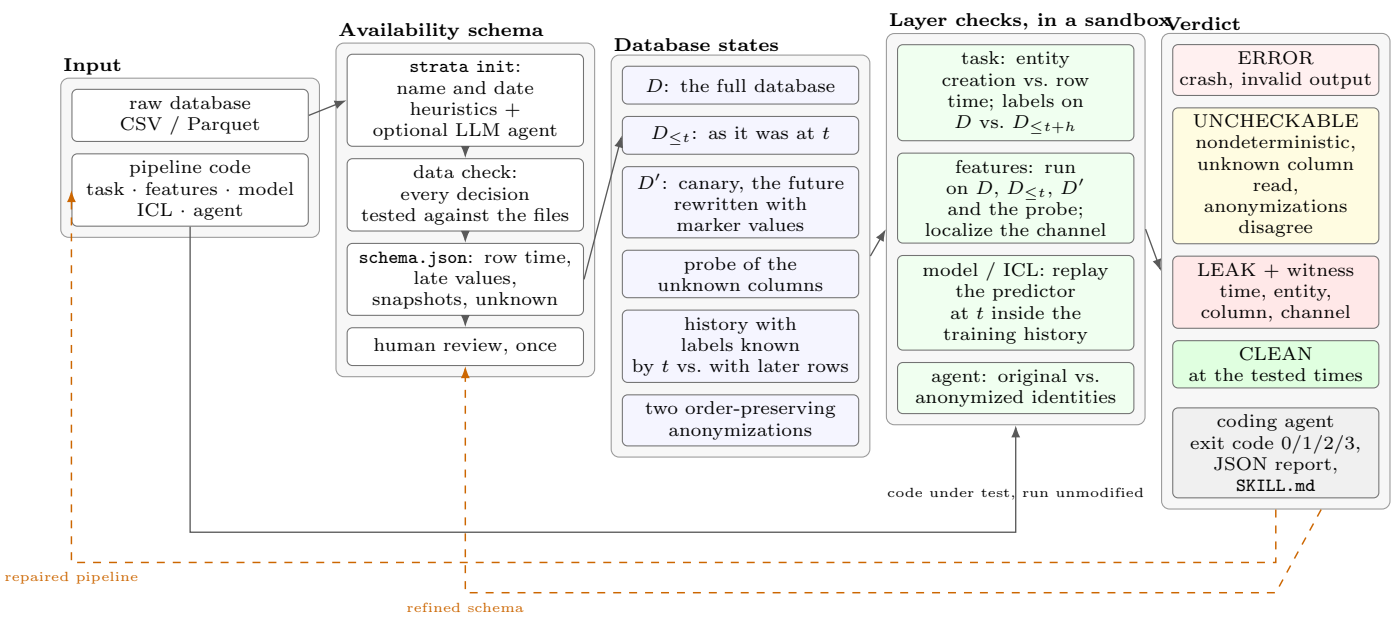

Input
raw database
CSV / Parquet
pipeline code
task · features · model
ICL · agent
Availability schema
strata init:
name and date
heuristics +
optional LLM agent
data check:
every decision
tested against the files
schema.json: row time,
late values,
snapshots, unknown
human review, once
Database states
D: the full database
$D_{\le t}$: as it was at t
D': canary, the future
rewritten with
marker values
probe of the
unknown columns
history with
labels known
by t vs. with later rows
two order-preserving
anonymizations
Layer checks, in a sandbox
task: entity
creation vs. row
time; labels on
D vs. $D_{\le t+h}$
features: run
on D, $D_{\le t}$, D'
and the probe;
localize the channel
model / ICL: replay
the predictor
at t inside the
training history
agent: original vs.
anonymized identities
Verdict
ERROR
crash, invalid output
UNCHECKABLE
nondeterministic,
unknown column
read,
anonymizations
disagree
LEAK + witness
time, entity,
column, channel
CLEAN
at the tested times
coding agent
exit code 0/1/2/3,
JSON report,
SKILL.md
code under test, run unmodified
repaired pipeline
refined schema